

Representing the Sliding-Tile Puzzle as Data

The sliding-tile board's data structure is just a list or array of integers. What makes it representative of an actual puzzle board is the way it's used by the functions in the program. The `displayBoard()`, `getNewBoard()`, `findBlankSpace()`, and other functions in this program all deal with this data structure.

Displaying the Board

The first function, `displayBoard()`, prints the board data structure on the screen. The Python code for the `displayBoard()` function is as follows:

Python

```
def displayBoard(board):
    """Display the tiles stored in `board` on the screen."""
    for y in range(SIZE): # Iterate over each row.
        for x in range(SIZE): # Iterate over each column.
            if board[y * SIZE + x] == BLANK:
                print('__ ', end='') # Display blank tile.
            else:
                print(str(board[y * SIZE + x]).rjust(2) + ' ', end='')
        print() # Print a newline at the end of the row.
```

The JavaScript code for the `displayBoard()` function is as follows:

```
function displayBoard(board) {
    // Display the tiles stored in `board` on the screen.
    document.write("<pre>");
    for (let y = 0; y < SIZE; y++) { // Iterate over each row.
        for (let x = 0; x < SIZE; x++) { // Iterate over each column.
            if (board[y * SIZE + x] == BLANK) {
                document.write('__ '); // Display blank tile.
            } else {
                document.write(board[y * SIZE + x].toString().padStart(2) + " ");
            }
        }
        document.write("<br />");
    }
}
```
